DAY 30 | HANDS-ON ACTIVITIES

LLM Evaluation and QA: Problem Statements

Seven problems that turn evaluation into automatic quality checks. All run offline; with a key, model-graded checks become meaningful.

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How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: Golden Set v1

Level: Warm-up | **Concepts:** dataset curation

Write 12 golden items in JSONL with id, question, expected, relevant chunk ids, tags, must_include and must_not_include. Save as golden_v1.jsonl.

Sample run / expected result

```
{"id": "leave-003", "question": "When does carried over leave expire?", "must_include": ["31
```

Hints

- must_not_include catches common wrong answers.

Problem 2: Deterministic Tests

Level: Easy | **Concepts:** PyTest parametrize

Write a parametrized PyTest that checks must_include and must_not_include for every golden item, with item ids shown in the test names.

Sample run / expected result

```
12 passed in 0.8s
```

Hints

- ids=lambda i: i["id"] gives readable test names.

Problem 3: Catch a Regression

Level: Medium | **Concepts:** prompt and config versions

A teammate changes k from 3 to 1 "to save cost" in config v2. Run the golden tests for v1 and v2 and report which items regressed.

Sample run / expected result

```
v1: 8/12 passed (offline)
v2: 6/12 passed
Regressed: leave-003, travel-003
```

Hints

- Compare per item, not only totals.

Problem 4: Context Precision and Recall

Level: Medium | **Concepts:** retrieval metrics

Compute precision, recall and average precision for k=1, 3 and 5 on the golden set. Explain the trade-off you see.

Sample run / expected result

```
k=1: precision 0.58 recall 0.58
k=5: precision 0.18 recall 0.92
```

Hints

- More chunks usually raise recall and lower precision.

Problem 5: BLEU from Scratch

Level: Medium | **Concepts:** n-grams, brevity penalty

Implement BLEU (1 to 4-grams, brevity penalty). Score an exact copy, a correct paraphrase and a wrong answer that reuses words. Explain the result.

Sample run / expected result

```
copy 1.00 | paraphrase 0.11 | wrong but similar words 0.63
```

Hints

- Add 1 smoothing so a missing 4-gram does not make the score zero.

Problem 6: Quality Gate Script

Level: Challenging | **Concepts:** thresholds, exit codes, CI

Write quality_gate.py that runs the golden set, checks recall >= 0.85 and that pass rate has not dropped more than 0.05 versus the v1 baseline, prints a report and exits with code 1 on failure. Show it fail for v2.

Sample run / expected result

```
v1: PASS (recall 0.92)
v2: FAIL (recall 0.58, pass rate dropped 0.17 vs baseline) -> exit 1
```

Hints

- sys.exit(1) makes the CI step fail.

Problem 7: Gate in the Pipeline

Level: Stretch | **Concepts:** Azure Pipelines

Add an Evaluate stage to your azure-pipelines.yml that runs the quality gate before the Deploy stages, and publishes the report as an artifact.

Sample run / expected result

```
Stages: Test -> Evaluate -> Build -> DeployDev -> DeployProd
```

Hints

- Store the API key as a secret pipeline variable, never in YAML.

Kihon (基本): the foundation